Decoding Left vs Right Hand Motor Imagery from a Two-Channel EEG Headset

A methods study, a comparative evaluation, and a route forward

Project folder `D:\trainsvm` | 8 October 2026

PART I - BACKGROUND, IN PLAIN LANGUAGE

1.1 What we are trying to do

When you imagine moving your left hand, the part of your brain that controls it becomes active. That part sits on the RIGHT side of the head, because the brain is crossed over. An electrode placed there - called C4 in the standard naming scheme - can pick up the change. For the right hand, the relevant electrode is C3 on the left side.

The change itself is a weakening, not a strengthening. At rest, the sensorimotor cortex produces a steady background rhythm: the mu rhythm at 8-13 Hz and the beta rhythm at 13-30 Hz. Think of it as an engine idling. When that patch of cortex gets busy - whether you move, or merely imagine moving - the idling stops and the rhythm gets quieter. This is called event-related desynchronisation, or ERD.

So the textbook signature is:

```
imagine LEFT hand   ->  rhythm WEAKENS at C4 (right side of head)
imagine RIGHT hand  ->  rhythm WEAKENS at C3 (left side of head)
```

A classifier's job is to look at the power at C3 and C4 in those bands and decide which hand was imagined.

In practice the effect is small, it varies enormously between people, and for a substantial minority of people it is absent altogether. That last fact has a name and we return to it in section 5.4.

1.2 What a classifier actually sees

The classifier never sees raw EEG. It sees FEATURES: a handful of numbers computed from each trial. The art is almost entirely in choosing those numbers.

The most useful single object is the COVARIANCE MATRIX of a trial. For a 2-channel recording it is a 2x2 table:

```
          C3        C4
C3   [ var(C3)   cov(C3,C4) ]
C4   [ cov(C3,C4)  var(C4)  ]
```

Its diagonal is the band power in each channel - exactly the quantity ERD changes. Its off-diagonal says how the two channels move together. Because the matrix is symmetric, a 2-channel recording gives **three independent numbers per trial per frequency band**. That is the whole information budget of this headset, and it explains a result that recurs throughout this study: the choice of classifier matters far less than people expect, because every classifier is reading the same three numbers.

1.3 The methods, and what each one adds

Log band power. Take the logarithm of the diagonal. Two numbers. The simplest thing that could possibly work.

CSP (Common Spatial Patterns). Instead of looking at C3 and C4 separately, find the weighted MIXTURE of channels whose power differs most between the two classes - one mixture tuned to be large for left and small

for right, another the reverse. With many electrodes this is powerful, because it can effectively synthesise a virtual electrode sitting exactly over the hand area. With 2 channels it can only produce 2 components, so it amounts to a rotation of the same information.

FBCSP (Filter-Bank CSP). Run CSP separately in several narrow frequency bands (say 4-8, 8-12, 12-16 Hz and so on), then select the most informative outputs. The motivation is that the exact frequency of a person's mu rhythm varies - one person peaks at 10 Hz, another at 12. A filter bank lets the method find each person's band rather than assuming one.

Riemannian tangent space. Covariance matrices do not live in ordinary flat space. They live on a curved surface (a manifold), and the straight-line average of two covariance matrices is not the natural average of them. The tangent-space method projects the matrices onto a flat space laid against the curved surface at the geometric mean, after which ordinary linear classifiers become valid. On many-channel data this is usually the strongest classical approach.

SVM (Support Vector Machine). The classifier itself. Given features, it finds the boundary separating the two classes with the widest possible margin. Its main setting is C: large C fits the training data tightly and risks memorising noise, small C gives a smoother boundary.

1.4 The two ideas that decide whether any of this works

Everything above is standard. The two ideas below are where the real gains turned out to be, and both are about the DATA rather than the model.

Idea 1: honest evaluation. To know how good a model is you must test it on data it did not learn from, and you must not make ANY decision - which band, which classifier, which settings - by looking at the test score. Two specific traps bit this project:

- *Leakage*: pieces of the same recording appearing in both training and test,

```
so the model partly recognises data it has seen.
```

- *Confounding*: something other than left/right also differs between the

```
classes. If every left trial was recorded on Monday and every right trial on
Tuesday, a model can score beautifully by learning the day.
```

Idea 2: alignment. EEG amplitude varies hugely between people, sessions and devices - electrode contact, impedance, hair, amplifier gain. A decision boundary fitted at one scale is meaningless at another. Euclidean Alignment (He & Wu, 2018) fixes this: for each recording session, compute the average covariance R over its trials, then transform every covariance

```
C  ->  R^(-1/2) C R^(-1/2)
```

After this each session's own average becomes the identity matrix, so all sessions arrive at the classifier in a common frame. Critically it uses **no labels**, so it can be applied to a new user's recording before you know what they were imagining. It needs no calibration and it is not cheating.

PART II - WHAT WE DID, AND WHAT HAPPENED

2.1 Stage one: the acute-stroke dataset

The project began with Liu et al. (2024), a published dataset of 50 acute stroke patients, 40 motor-imagery trials each, 30 electrodes. The paper reports up to 72.21 % accuracy.

We could not reproduce it. Re-implementing the authors' own evaluation procedures and running each one twice - once on the true labels, once on randomly shuffled labels - showed why:

procedure (from the authors' code)	true labels	SHUFFLED labels	paper
FBCSP + SVM	48.27 %	47.05 %	57.57 %
TSLDA + FgMDM "fusion"	54.02 %	47.95 %	61.20 %
TWFB + FgMDM (their best)	65.95 %	**63.79 %**	72.21 %

A method that scores 63.79 % on labels containing no information is not measuring the brain. Reading `TWFB_DGFMDM.m` explains it: the procedure tries eight frequency bands, each on a fresh random split, and keeps the band with the highest TEST accuracy. That is selecting on the test set, and the more options you try the higher the luckiest one scores.

We then ran nine models through one honest protocol - identical folds for every model, every choice made by inner cross-validation on training data only:

CSP + LDA	50.45 %
FBCSP + SVM	52.70 %
Tangent space + SVM	53.30 %
TSLDA / FgMDM / fusion	53.50 %
TWFB + FgMDM	55.40 % <- best honest result

Only about 12 of 50 patients could be decoded above chance, and it was the same patients every time. **The ceiling of this dataset is roughly 55 %**, and no feature set or classifier moves it. In parallel we ran sixteen configurations of hand-crafted features (mean, median, skewness, kurtosis of the band energy at C3 and C4, plus the C3/C4 power ratio). They spanned 0.526 to 0.539 balanced accuracy. A great deal of effort moved the number by thirteen thousandths, underneath a ceiling of 0.554.

Lesson: when a long sequence of model changes produces no movement, the problem is upstream of the model.

2.2 Stage two: changing the dataset

We switched to BCI Competition IV dataset 2b, which suits this project far better:

- 9 healthy subjects, 5 sessions each on different days, ~6,520 trials total
- exactly two classes, left vs right hand, perfectly balanced
- **three bipolar channels: C3, Cz, C4** - essentially our montage
- **250 Hz** - our headset's native rate, so no resampling anywhere
- cues randomised WITHIN each session, so class is not confounded with session
- a deliberately minimal montage, chosen by its designers to emulate portable

```
BCI scenarios
```

The pipeline: band-pass 8-30 Hz, epoch 0.5-3.5 s after the cue, one covariance matrix per trial, optional Euclidean Alignment, then one of seven models. Two evaluation protocols:

- **P1, within-subject**: train on that person's sessions 1-3, test on 4-5.
- **P2, cross-subject**: train on 8 subjects, test on the 9th. This is the

```
structural analogue of what we actually want - train on public data, use on
a new person.
```

2.3 The comparative study

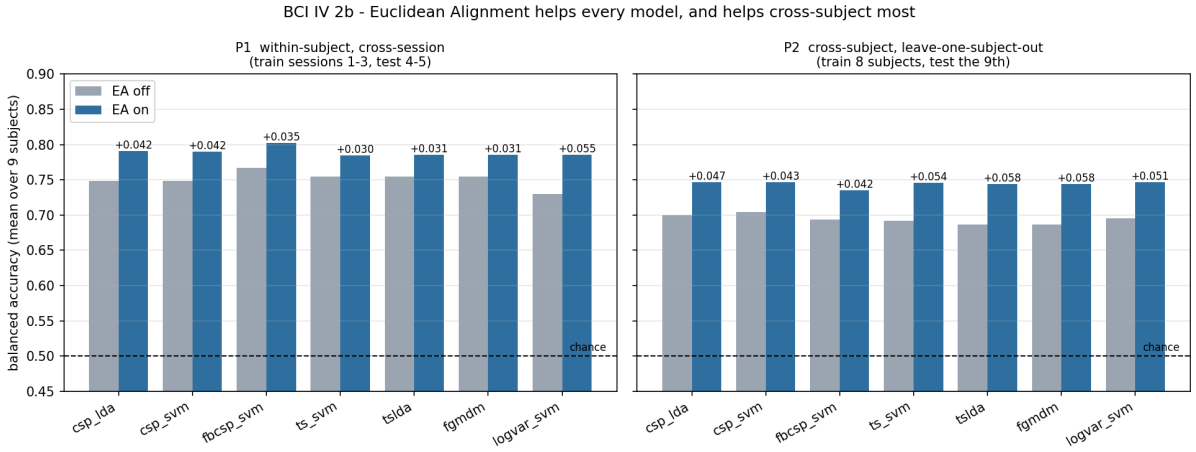


Figure 1. Euclidean Alignment improves every model under both protocols, and improves cross-subject transfer most. Numbers above the bars are the gain from alignment.

Balanced accuracy, mean over 9 subjects. Chance = 0.500, and because the classes are exactly balanced there is no majority-class trap in this table.

P1 - within-subject (train sessions 1-3, test 4-5)

model	EA off	EA on	gain from EA
CSP + LDA	0.7480	0.7905	+0.0425
CSP + SVM	0.7480	0.7898	+0.0418
FBCSP + SVM	0.7668	**0.8020**	+0.0352
Tangent + SVM	0.7548	0.7846	+0.0298
TSLDA	0.7544	0.7855	+0.0311
FgMDM	0.7544	0.7855	+0.0311
Log variance	0.7299	0.7851	+0.0552

P2 - cross-subject, leave-one-subject-out

model	EA off	EA on	gain from EA	subjects > chance
CSP + LDA	0.6996	**0.7468**	+0.0472	9/9
CSP + SVM	0.7039	0.7468	+0.0429	9/9
FBCSP + SVM	0.6936	0.7353	+0.0417	9/9
Tangent + SVM	0.6918	0.7456	+0.0538	9/9
TSLDA	0.6864	0.7440	+0.0576	9/9
FgMDM	0.6864	0.7440	+0.0576	9/9
Log variance	0.6953	**0.7468**	+0.0515	9/9

The three levers, measured

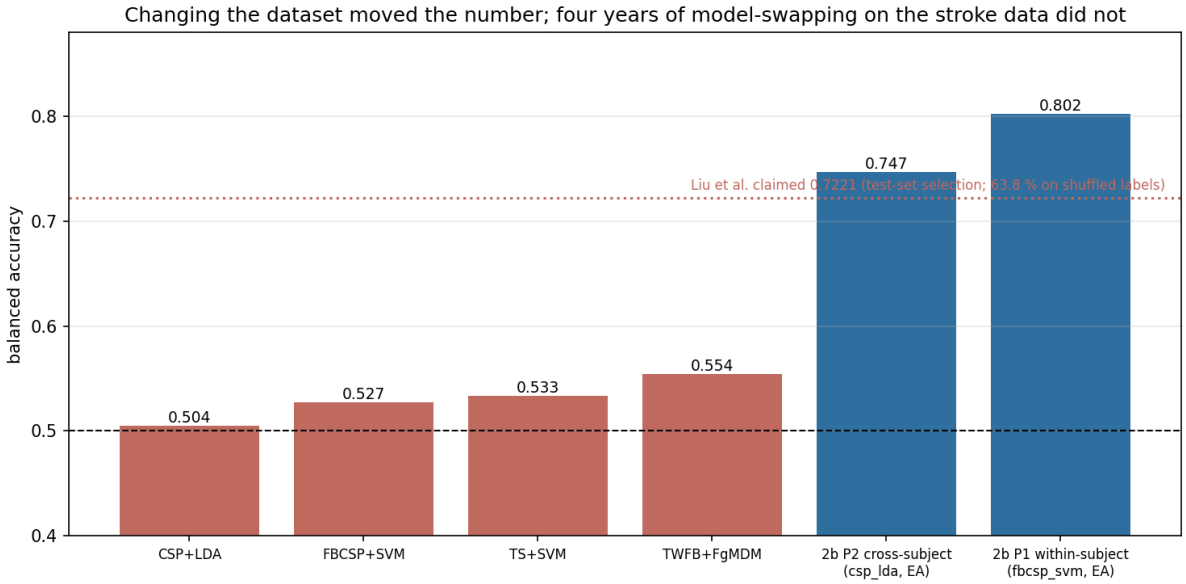


Figure 2. Changing the dataset moved the result; four years of model-swapping on the stroke data did not. The dotted line is the accuracy claimed by Liu et al., which scores 63.8% on shuffled labels.

```
choice of dataset        ~0.19     stroke 0.554  ->  2b 0.747
Euclidean Alignment      +0.045    positive in 28 of 28 configurations
choice of model          ~0.003    the entire P2 spread with EA on
```

Euclidean Alignment helped in every single one of 28 configurations: mean +0.045, smallest +0.017, largest +0.069. The choice between filter-bank CSP and three log-variance numbers was worth 0.0115, and log variance tied for best cross-subject. This is the 2x2-covariance argument from section 1.2 showing up in the measurements.

2.4 A bug worth describing

The first run of this experiment returned 0.49-0.51 for every model, every protocol, every alignment setting - 28 configurations sitting exactly on chance.

The project's plan document, written BEFORE the run, had said: *"P1 should land around 75-80 %. If it comes out near 55 %, something is wrong with my epoching - most likely the cue index convention."*

It was the cue index convention. The `trial` field in the 2b files indexes the TRIAL START; the paradigm is 3 s of fixation, then the cue, then 4 s of imagery. Our window sat entirely inside the fixation period, where by construction there is nothing to classify.

A diagnostic that assumes neither convention - slide a 2 s window across the trial and measure how well C3-minus-C4 power separates the classes - located it immediately:

```
offset after `trial`:  0.0s  0.5s  1.0s  1.5s  2.0s  2.5s  3.0s  3.5s  4.0s
AUC:                  0.493 0.496 0.505 0.497 0.425 0.339 0.263 0.212 0.228
                      <------ chance ------>              cue   peak
```

(AUC below 0.5 is just the sign of the feature: right-hand imagery desynchronises C3, so C3-minus-C4 power goes DOWN. That is the textbook direction, measured.)

This matters as methodology, not just as a fix. A table of chance results is a perfectly plausible-looking negative finding - "three channels is not enough" would have been believed. Writing the expected range down in advance is the only reason it was read as a bug.

2.5 Stage three: testing our own headset

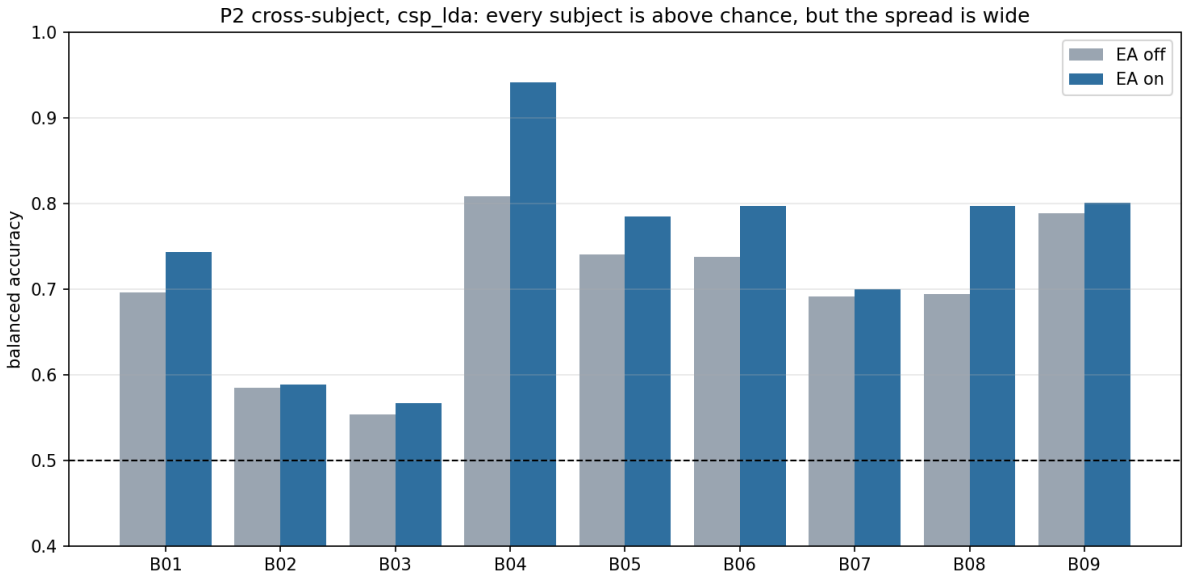


Figure 3. Cross-subject results per subject. Every subject is above chance, but the spread is wide: B03 at 0.57 against B04 at 0.94. BCI illiteracy is visible in the data.

Our recordings: a Gelo BCI headband, 4 live electrodes of 8, ch2 and ch3 taken as C3 and C4, 250 Hz, 15 files, 72 usable epochs.

First, what does dropping Cz cost? The 2b model was retrained on C3+C4 only:

```
model           3 channels    2 channels    cost
CSP + LDA         0.7468        0.7441    -0.0027
Tangent + SVM     0.7456        0.7439    -0.0017
FgMDM             0.7440        0.7442    +0.0002
Log variance      0.7468        0.7442    -0.0026
```

Still 9/9 subjects above chance. **Two electrodes are as good as three.** The channel count is not the obstacle.

Then the transfer. Trained on all nine 2b subjects, tested on our 72 epochs with the identical window and band. Because each of our files holds a single class, the honest unit is the file - 15 decisions, not 72.

model	EA	epoch bal.	file bal.	files correct	permutation p
CSP + LDA	off	0.486	0.450	6/15	0.832
CSP + LDA	on	0.480	0.300	5/15	0.983
Tangent	off	0.447	0.400	6/15	0.897
Tangent	on	0.490	0.250	4/15	0.993
FgMDM	on	0.490	0.250	4/15	0.993
Log var	on	0.500	0.350	6/15	0.957

Everything is at chance. Best permutation p is 0.832 over 10,000 exact file-label shuffles.

Because every configuration landed slightly BELOW chance, we tested one pre-specified physical alternative - that ch2 and ch3 are swapped - with a Bonferroni-corrected threshold of $p < 0.025$. Best result anywhere: 0.500 at $p = 1.000$. Swapping does not rescue it.

A descriptive check without any classifier makes the situation clearest. AUC of log(C3) - log(C4) for predicting right-hand imagery:

```
2b        0.2174     |AUC - 0.5| = 0.283   textbook contralateral ERD
our data  0.5127     |AUC - 0.5| = 0.013   twenty times weaker
```

Why this negative result is strong. An earlier report in this project had to say of its own test: *"the models never worked on the training data, so their behaviour on our data cannot be evidence about our data. A broken ruler does not measure a table."* That is no longer the situation. The same model, same code, same window, same band reaches 0.744 on a person it has never seen, with 9 of 9 subjects above chance - and then reaches

chance on ours. The failure is isolated to the recordings.

One genuine improvement is worth recording: predictions came out 39-50 % "right". In the earlier experiment, absolute-energy models predicted "right" for 85-100 % of epochs because our recordings sit 3.1x higher in energy and every epoch fell on one side of a boundary fitted to another scale. **Euclidean Alignment removed that collapse entirely.** The classifier is now genuinely undecided rather than saturated - undecided because there is nothing to decide.

2.6 Stage four: how much calibration data would a new user need?

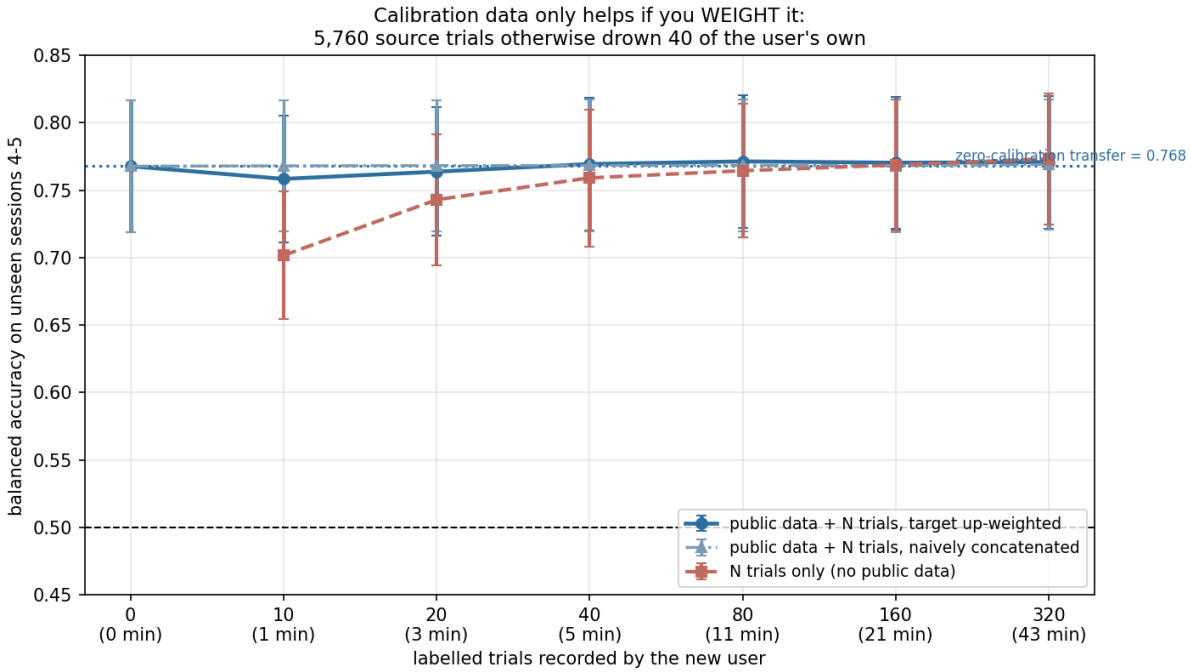


Figure 4. Calibration data buys very little once Euclidean Alignment is applied, and naive concatenation wastes it entirely. The public dataset is worth roughly 160 trials of the user's own time.

If a new user records some labelled trials of their own, how fast does accuracy climb? We measured this on 2b: start from the model trained on 8 other subjects, add N labelled trials from the target, test on the target's held-out sessions. Each N repeated 10 times with different random draws.

trials	minutes	public data + N, concatenated	+ N, up-weighted	N only, no public data
0	0	0.7678	0.7678	--
10	1	0.7681	0.7584	0.7017
20	3	0.7683	0.7637	0.7430
40	5	0.7684	0.7694	0.7591
80	11	0.7684	0.7713	0.7644
160	21	0.7681	0.7704	0.7688
320	43	0.7691	0.7708	0.7733

Three findings, two of them surprising:

1. **Calibration barely helps.** 320 of the user's own trials - 43 minutes in

the chair - moves accuracy by less than 0.01. Once Euclidean Alignment is in place, **the alignment does the work that calibration would otherwise do.** This is good news practically: a zero-calibration decoder is viable.

2. **Naive concatenation wastes calibration data entirely** (+0.001 at 320

trials). 5,760 source trials simply drown 320 target trials. Up-weighting the target to half the training mass recovers a little (+0.004 at 80 trials) but not much. If anyone reports large gains from "adding calibration data", check how they weighted it.

3. **The public dataset is worth about 160 trials - 21 minutes - of the user's own time**, which is where the calibration-only curve crosses the transfer line.

PART III - WHAT THE LITERATURE SAYS WOULD HELP

We surveyed current methods against this project's specific constraints: two channels, a consumer headset, one user, and a small dataset.

3.1 Deep learning

EEGNet is a compact convolutional network widely used for BCI. The honest finding across the literature is that **it does not beat classical methods on motor imagery**. In its own paper, on the sensorimotor-rhythm task, EEGNet-8,2 achieved performance comparable to the established FBCSP method and ShallowConvNet. Independent comparisons agree, with one noting it is interesting that FBCSP with regularised LDA performs motor-imagery classification as well as EEGNet even though it is not deep learning.

Analysis of what EEGNet learns explains why: comparison with FBCSP revealed significant correlations between EEGNet's learned spatial filters and FBCSP's optimal spatial patterns, suggesting the network implicitly rediscovers the same variance-maximising features. On 22-channel data it rediscovers CSP. On 2 channels there is almost nothing to rediscover.

Verdict for this project: not worth it. It would add large complexity, a GPU dependency and substantial overfitting risk to recover features we already compute in closed form.

3.2 Data augmentation

The most thorough study compared 13 augmentations across two tasks, datasets and models, and found that adequate augmentation can bring up to 45 % accuracy improvement in low-data regimes - but also that there is no single best augmentation strategy, as the good augmentations differ by task. A later study on 12 augmentation methods confirmed the double edge: augmentation can significantly affect classification accuracy, with some methods degrading performance while others improve generalisation.

Verdict: relevant only once a deep model is in use, and only with careful validation. Augmentation addresses overfitting. Our classical pipeline with three features per trial is not overfitting - it is reading data that contains no signal. Augmenting noise produces more noise.

3.3 Transfer learning and alignment

Already implemented, and the single largest methodological gain in this project. Euclidean Alignment aligns EEG trials directly in Euclidean space, so any downstream processing can be applied to the aligned trials; its computational cost is very low; and it is unsupervised, needing no label information from the new subject. It outperformed Riemannian alignment and several no-alignment approaches, and generally improves decoding accuracy especially for shared cross-subject models.

One caveat for future work: aligning across datasets recorded with different acquisition systems, electrode numbers and positions remains an open problem, because those differences introduce heterogeneous feature spaces. Our 2b-to-Gelo transfer is exactly that harder case.

3.4 Self-supervised pretraining (JEPA and EEG foundation models)

Joint Embedding Predictive Architectures learn by predicting latent representations rather than reconstructing the raw signal. Signal-JEPA was built specifically for cross-dataset transfer, with a domain-specific spatial block masking strategy, evaluated on 54 subjects across motor imagery, ERP and SSVEP. Its central conclusion is directly relevant: the results highlight the importance of spatial filtering for accurate downstream classification.

Related work includes EEG-VJEPA, which models multi-channel EEG as spatiotemporally masked sequences and reaches state-of-the-art on the TUH Abnormal dataset, and Laya, a LeJEPA-based EEG foundation model. The motivation behind Laya is itself a warning about the whole family: reported improvements of EEG foundation models over smaller task-specific models are often modest, sensitive to downstream adaptation

and fine-tuning strategy, and limited under linear probing.

Verdict: a legitimate future-work section, not a route to accuracy here. These models are pretrained on tens of thousands of sessions. We have 6,520 training trials and 72 test epochs, on two channels.

3.5 Reinforcement learning

RL needs a reward signal and a loop to close. On a fixed offline dataset with known labels, the only available reward IS the label, so an RL agent collapses into a higher-variance way of fitting the same boundary.

Where RL genuinely applies is the closed loop: error-related potentials combined with RL, where the framework learns the user's intention directly from brain signals and uses the neural signature of errors as the learning signal. That requires a live session, a feedback display, and a second decoder for the ErrP itself. Note the same study's warning: the use of motor imagery in a high-speed interaction paradigm proved ineffective for most participants.

Verdict: not applicable offline. Potentially relevant once there is a working online system - which is several steps away.

3.6 Co-adaptive calibration - the most relevant finding

This is the single most actionable item in the literature for this project, and it builds directly on what we have already built.

Vidaurre et al.'s machine-learning-based co-adaptive calibration **starts with a subject-independent classifier** - exactly the artefact produced by this study -

and evolves it to a subject-optimised state within one session while the user interacts continuously. Early runs use supervised co-adaptive learning of user and machine; later runs use unsupervised adaptation to track feature drift. With this approach, and **without any offline calibration**, six users including one novice obtained good performance after 3 to 6 minutes of adaptation. More importantly, the guided learning allowed participants with BCI illiteracy to gain significant control in under 60 minutes.

The hypothesis motivating it is that one main difficulty for a BCI user is the transition from offline calibration to online feedback - which is precisely the transition our recordings never made.

3.7 BCI illiteracy and user training

A non-negligible portion of users cannot control a motor-imagery BCI at all. The commonly cited range is 15-30 %, and the phenomenon is generic to all motor-imagery-based BCIs. Our own data shows it: two of nine 2b subjects (B02 at 0.57, B03 at 0.52) sit near chance even when trained on their own data.

It is not necessarily permanent. Feedback training has shown that subjects previously categorised as BCI illiterate could gain BCI control within one session, including one subject who had no sensorimotor idle rhythm at the start and developed one through feedback training. Other routes under investigation include hybrid somatosensory stimulation training aimed specifically at poorly performing users.

Verdict: directly relevant, and currently unaddressed. All our recordings were made without any feedback. The user has never seen whether their imagery was being decoded correctly, and so has had no opportunity to learn.

PART IV - RECOMMENDATIONS

In descending order of expected effect.

4.1 Re-record with a corrected protocol [essential, blocks everything else]

Nothing measured on the current recordings can mean anything, because class is perfectly confounded with session. Copy 2b's design, which is known to work:

- **Interleave left and right randomly within every session.** This is the

`single most important change. As long as one file holds one class, any difference between recording sessions reproduces as apparent accuracy.`

- **Software-timed cues**, with the marker written by the program. Current

`trials are self-clicked, which means a real hand movement at t=0.`

- **Fixed-length trials**: 2 s rest, cue, 4 s imagery. Current trials run

`from 4.9 s to 148 s.`

- **At least 40 trials per class per session**, ideally 80, across several

`sessions on different days.`

- **Record the electrode positions** and photograph the placement.
- Keep a rest block in every session as a control.

4.2 Add online feedback [the largest untried lever]

Every recording so far has been open-loop. The user has never been shown whether their imagery was decoded correctly. Given that feedback training can bring previously BCI-illiterate users into control within a session, and that the offline-to-online transition is itself identified as the main difficulty, this is likely worth more than any remaining algorithmic change.

The decoder to put in the loop already exists: `gelo_svm_pipeline.py`, trained on 2b, requires no calibration, and is validated at 0.744 on unseen people.

4.3 Deploy co-adaptive calibration [follows directly from 4.2]

Start each session with the subject-independent model, adapt it supervised for the first few minutes while the user receives feedback, then switch to unsupervised adaptation to track drift. This is the published recipe, it needs no offline calibration session, and our calibration-curve experiment independently shows that a static offline calibration would have bought under 0.01 anyway.

4.4 Verify electrode placement [cheap, and currently unknown]

The left/right ORDER has been tested and is not the problem. What remains untested is whether the two electrodes sit over the hand area at all. 2b uses bipolar pairs placed about 2.5 cm either side of C3 and C4; a headband sitting too far forward or back records a different rhythm entirely. Also still unknown: what ch1 and ch4 are. If either is a usable scalp site, the montage grows.

4.5 Do not pursue, and why

- **A better classifier.** Worth ~0.003 on two channels. Measured.
- **More features.** Sixteen configurations of mean/median/skewness/kurtosis

`spanned 0.013 on the stroke data. Measured.`

- **Deep learning.** Comparable to FBCSP on motor imagery, at much greater

`cost and overfitting risk.`

- **Data augmentation.** Addresses overfitting, which is not our problem.
- **Offline calibration sessions.** Under 0.01 once alignment is applied.

`Measured.`

PART V - WHAT TO TAKE AWAY

On the science. Left-vs-right motor imagery is decodable from two electrodes. We measured 0.744 balanced accuracy on a person the model had never seen, with all nine test subjects above chance, using C3 and C4 alone. The limiting factor is not the number of electrodes, not the features, and not the classifier.

On the method. Three practices earned their keep, each by catching a real error:

1. *Write the expected result down before running.* It is what turned a `plausible-looking negative result into a located bug.`
2. *Test every procedure on shuffled labels.* It is what revealed that a `published 72 % scores 63.8 % on random labels.`
3. *Score at the level of the independent unit.* 72 epochs from 15 `single-class files is 15 decisions.`

On the project. The decoder is built, validated and ready. What it needs is data recorded in a way that can be measured - interleaved classes, timed cues, fixed trials - and a feedback loop so the user can learn. Both are recording decisions, not modelling decisions.

APPENDIX A - FILES

```
gelo_svm_pipeline.py    the complete decoder, written to be read
00_PLAN.md              design and predictions, written before any run
s00_diagnose_window.py  locates the imagery window without assuming it
s01_fetch_2b.py         download, epoch, cache covariances
s02_run_models.py       7 models x 2 alignments x 2 protocols
s03_report.py           figures
s04_gelo_transfer.py    2-channel ceiling, and the transfer to our data
s05_polarity_check.py   the channel-swap test
s06_calibration_curve.py how many trials a new user needs
REPORT.md               full technical report with all numbers
HOW_TO_RUN.md           reproduction instructions
results\                every CSV, log and figure
```

APPENDIX B - REFERENCES

He & Wu (2018). Transfer Learning for Brain-Computer Interfaces: A Euclidean

Space Data Alignment Approach. arXiv:1808.05464

Lawhern et al. (2018). EEGNet: A Compact Convolutional Network for EEG-based

Brain-Computer Interfaces. arXiv:1611.08024

Guetschel, Moreau & Tangermann (2024). S-JEPA: towards seamless cross-dataset

transfer through dynamic spatial attention. arXiv:2403.11772

Rommel, Paillard, Moreau & Gramfort (2022). Data augmentation for learning

predictive models on EEG: a systematic comparison. arXiv:2206.14483

Vidaurre et al. Machine-learning-based co-adaptive calibration for

brain-computer interfaces.

Xavier Fidencio et al. (2025). Error-related Potential driven Reinforcement

Learning for adaptive Brain-Computer Interfaces. arXiv:2502.18594

Panchavati et al. Laya: A LeJEPA Approach to EEG via Latent Prediction over

Reconstruction. arXiv:2603.16281

Leeuwis et al. (2021), via Park, Ha, Kim & Kim (2021). Improving Motor

Imagery-Based BCI Performance Based on Sensory Stimulation Training.
Front. Neurosci. 15:732545

Liu et al. (2024). An EEG motor imagery dataset for brain computer interface

in acute stroke patients. Scientific Data 11:131

BCI Competition IV dataset 2b (Leeb et al.), BNCI Horizon 2020 id 004-2014